



**RESEARCH / PROJECT ETHICS REVIEW FORM**  
**For Thesis / Capstone / Dissertation Proposals**  
**(ver.Oct2024c)**

**Names of Student Researcher(s):**

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**Russell Emmanuel Galan**

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**Research/Thesis/Capstone Title:** A New Rendering and Plume Simulation Framework for Visualizing Taal Volcano Eruption Plumes

**College:** College of Computer Studies

**Department:** Software Technology

**Course:** BS Computer Science major in Software Technology

**Expected Duration of the Project:** from: Aug 6, 2026 to Aug 6, 2027

**PROTOCOL:**

*(To be filled up by proponents; Use the Ethics Checklists as guides in determining areas for ethical concern/consideration. Attach and properly reference the accomplished ethics checklists and supporting documents.)*

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Reference Document<br>(include page number) |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| Target participants         | <p><b>Sample Size:</b> 10 to 20 voluntary participants.</p> <p><b>Inclusion Criteria:</b> Participants must be computer science students or developers with a relevant background in 3D graphics or software development.</p> <p><b>Exclusion Criteria:</b> Individuals without a relevant computer science, software development, or 3D graphics background.</p> <p><b>Withdrawal Criteria:</b> Participants may exit the software at any stage and can choose to withdraw from the evaluation without academic or personal penalty.</p> | Scope and Limitations<br>(page 4-5)         |
| Data to be used / collected | <p><b>Technical Data:</b> Empirical hardware performance logs including Frame Rate (FPS), CPU/GPU resource utilization percentage, and frame rendering times (ms).</p>                                                                                                                                                                                                                                                                                                                                                                    | Scope and Limitations<br>(page 4-5)         |

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|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
|                 | <p><b>Human Data:</b> Quantitative usability scores via the System Usability Scale (SUS) questionnaire and qualitative feedback regarding visual artifacts or rendering anomalies. No personally identifiable information (PII) such as names, student numbers, or email addresses will be tied to the responses. Also, feedback gathered from a closing interview regarding suggestions for improving the interface and overall performance.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                  |
| Brief procedure | <ol style="list-style-type: none"> <li>1. <b>Screening:</b> Interested participants complete a short pre-screening form confirming they meet the inclusion criteria (collegiate CS student, graphics programmer, or software developer with prior exposure to real-time graphics or game development).</li> <li>2. <b>Consent:</b> Eligible participants are presented with the Informed Consent Form (ICF), which outlines the evaluation's scope, hardware interactions, session length, and data privacy measures.</li> <li>3. <b>Orientation/Briefing:</b> Participants are given a brief, neutral walkthrough of the interactive Taal Volcano Plume simulator's controls and features.</li> <li>4. <b>Usability Task:</b> Participants will use the software to recreate specific Taal Volcano eruption plumes in response to predefined scenarios.</li> <li>5. <b>Post-Session Questionnaire :</b> Participants will complete an anonymous, self-administered SUS questionnaire to rate the software's ease of use and compare visual consistency.</li> <li>6. <b>Feedback:</b> A closing interview will be conducted to gather suggestions for improving the interface and overall performance.</li> </ol> | Scope and Limitations (page 4-5) |

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| Potential risks                                                             | <p><b>Physical:</b> Minimal risks, though participants may experience minor eye strain or visual fatigue from interacting with the 3D graphics simulator.</p> <p><b>Technical:</b> Participants may experience frustration if they encounter bugs or usability issues with the experimental prototype.</p> <p><b>Mitigation:</b> Researchers will provide immediate technical support for major difficulties, and participants are encouraged to request breaks at any time if they feel tired or frustrated.</p>                                                                                                                                                                                                                                                                                                             |                                  |
| Applicable Terms of Use or Licensing Agreement of Data, Models and/or Tools | <p><b>AnitoPlume Source Code:</b> Used with explicit permission and under the supervision of the thesis adviser, Neil Del Gallego.</p> <p><b>Libraries:</b> Eigen (Mozilla Public License v2.0) and dearIMGUI (MIT License).</p> <p><b>Graphics Toolkits &amp; SDKs:</b> Khronos Vulkan SDK, Microsoft DirectX 12 SDK, and Khronos OpenGL specifications.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Scope and Limitations (page 4-5) |
| Steps to safeguard the participants and/or data                             | <p><b>Participant Safety:</b> Testing sessions will be kept brief to prevent eye strain. Participants will be explicitly informed via the ICF that participation is entirely voluntary and that they may withdraw from the usability test at any point without academic or personal penalty.</p> <p><b>Data Safety:</b> All performance logs and usability questionnaire responses will be fully anonymized using alphanumeric identifiers.</p> <p><b>Data Storage:</b> Survey sheets and digital spreadsheets will be secured in a password-protected cloud storage repository accessible only to the primary student researchers.</p> <p><b>Data Retention:</b> Anonymized data will be kept 4 to 5 years following final publication or degree conferral to allow for reproducibility audits before permanent deletion</p> | Scope and Limitations (page 4-5) |

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| Other concerns and considerations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | None at this point in time |                                           |
| <b>PANEL'S ASSESSMENT/COMMENTS:</b><br><br>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                            |                                           |
| <b>PANEL'S DECISION:</b><br><br><div style="margin-left: 20px;">             € <b>Approved</b> <i>(Proponent/s may proceed with the study/data collection)</i><br/>             € <b>Modifications Required</b> <i>(Proponent/s should address ethical issues, follow the panel's recommendations for required modifications, and resubmit Ethics Review Form for approval before proceeding with the study or before starting with data collection)</i><br/>             € <b>For Full Review</b> <i>(Proponent/s should submit the proposal to the University Research Ethics Review Committee (RERC) and obtain a research ethics clearance before proceeding with the study or before starting with data collection)</i> </div> |                            |                                           |
| <b>JUSTIFICATION:</b><br><div style="border-bottom: 1px solid black; height: 1.2em; margin-bottom: 5px;"></div> <div style="border-bottom: 1px solid black; height: 1.2em; margin-bottom: 5px;"></div> <div style="border-bottom: 1px solid black; height: 1.2em; margin-bottom: 5px;"></div> <div style="margin-left: 40px; margin-top: 20px;"> <div style="border-bottom: 1px solid black; height: 1.2em; margin-bottom: 5px;"></div> <div style="border-bottom: 1px solid black; height: 1.2em; margin-bottom: 5px;"></div> </div>                                                                                                                                                                                               |                            |                                           |
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| <b>Name and Signature of Adviser/Mentor</b><br><br>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                            |                                           |
| <b>Date:</b> <span style="border-bottom: 1px solid black; display: inline-block; width: 150px;"></span>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                            |                                           |
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| <b>Name and Signature of Panel Member</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                            | <b>Name and Signature of Panel Member</b> |

**Date:** \_\_\_\_\_

**Date:** \_\_\_\_\_

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**Name and Signature of Panel Member**

**Date:** \_\_\_\_\_